MEDICAL IMAGING PHYSICS

Fourth Edition

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Ad hoc, ad loc
and quid pro quo
so little time
so much to know.

Jeremy Hillary Boob, Ph.D.
The Nowhere Man in the Yellow Submarine

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PREFACE

Writing and rewriting a text such as *Medical Imaging Physics* over several editions presents two challenges. The first is to keep the information fresh and relevant. This is a particular challenge in medical imaging, because the field is evolving so rapidly. The third edition of this text was published in 1992, just 10 short years ago. Yet in that text no mention was made of topics such as photodiode or direct conversion digital x-ray imagers; digital mammography; digital fluoroscopy; power Doppler ultrasound; functional magnetic resonance imaging; elastography; or helical CT scanning. This is just a partial list of imaging approaches that must be covered today in any text of imaging physics. Being involved in a dynamic and rapidly changing field is one of the more enjoyable aspects of medical imaging. But it places heavy demands on authors trying to provide a text that keeps up with the field.

The second challenge is no less demanding than the first. That challenge is to keep the text current with the changing culture of how people learn, as well as with the educational experience and pedagogical expectations of students. These have changed remarkably over the 30 years since this book first appeared. For maximum effect, information today must be packaged in various ways, including self-contained segments, illustrations, highlights, sidebars, and examples and problems. In addition, it must be presented in a manner that facilitates learning and helps students evaluate their progress. Making the information correct and complete is only half the battle; the other half is using a format that helps the student assimilate and apply it. The latter challenge reflects not only today's learning environment, but also the tremendous amount of information that must be assimilated by any student of medical imaging.

In recognition of these challenges, the authors decided two years ago to restructure *Medical Imaging Physics* into a fourth edition with a fresh approach and an entirely new format. This decision led to a total rewriting of the text. We hope that this new edition will make studying imaging physics more efficient, effective, and pleasurable. It certainly has made writing it more fun.

Medical imaging today is a collaborative effort involving physicians, physicists, engineers, and technologists. Together they are able to provide a level of patient care that would be unachievable by any single group working alone. But to work together, they must all have a solid foundation in the physics of medical imaging. It is the intent of this text to provide this foundation. We hope that we have done so in a manner that makes learning enriching and enjoyable.

William R. Hendee, Ph.D.
E. Russell Ritenour, Ph.D.

PREFACE TO THE FIRST EDITION

This text was compiled and edited from tape recordings of lectures in medical radiation physics at the University of Colorado School of Medicine. The lectures are attended by resident physicians in radiology, by radiologic technologists and by students beginning graduate study in medical physics and in radiation biology. The text is intended for a similar audience.

Many of the more recent developments in medical radiation physics are discussed in the text. However, innovations are frequent in radiology, and the reader should supplement the book with perusal of the current literature. References at the end of each chapter may be used as a guide to additional sources of information.

Mathematical prerequisites for understanding the text are minimal. In the few sections where calculus is introduced in the derivation of an equation, a description of symbols and procedures is provided with the hope that the use of the equation is intelligible even if the derivation is obscure.

Problem solving is the most effective way to understand physics in general and medical radiation physics in particular. Problems are included at the end of each chapter, with answers at the end of the book. Students are encouraged to explore, discuss and solve these problems. Example problems with solutions are scattered throughout the text.

Acknowledgments

Few textbooks would be written without the inspiration provided by colleagues, students and friends. I am grateful to all of my associates who have contributed in so many ways toward the completion of this text. The original lectures were recorded by Carlos Garciga, M.D., and typed by Mrs. Marilyn Seckler and Mrs. Carolyn McCain. Parts of the book have been reviewed in unfinished form by: Martin Bischoff, M.D., Winston Boone, B.S., Donald Brown, M.D., Frank Brunstetter, M.D., Duncan Burdick, M.D., Lawrence Coleman, Ph.D., Walter Croft, Ph.D., Marvin Daves, M.D., Neal Goodman, M.D., Albert Hazle, B.S., Donald Herbert, Ph.D., F. Bing Johnson, M.D., Gordon Kenney, M.S., Jack Krohmer, Ph.D., John Pettigrew, M.D., Robert Siek, M.P.H., John Taubman, M.D., Richard Trow, B.S., and Marvin Williams, Ph.D. I appreciate the comments offered by these reviewers. Edward Chaney, Ph.D., reviewed the entire manuscript and furnished many helpful suggestions. Robert Cadigan, B.S., assisted with the proofreading and worked many of the problems. Geoffrey Ibbott, Kenneth Crusha, Lyle Lindsey, R.T., and Charles Ahrens, R.T., obtained much of the experimental data included in the book.

Mrs. Josephine Ibbott prepared most of the line drawings for the book, and I am grateful for her diligence and cooperation. Mrs. Suzan Ibbott and Mr. Billie Wheeler helped with some of the illustrations, and Miss Lynn Wisehart typed the appendixes. Mr. David Kuhner of the John Crerar Library in Chicago located many of the references to early work. Representatives of various instrument companies have helped in many ways. I thank Year Book Medical Publishers for encouragement and patience and Marvin Daves, M.D., for his understanding and support.

I am indebted deeply to Miss Carolyn Yandle for typing each chapter many times, and for contributing in many other ways toward the completion of the book.

Finally, I wish to recognize my former teachers for all they have contributed so unselfishly. In particular, I wish to thank Fred Bonte, M.D., and Jack Krohmer, Ph.D., for their guidance during my years as a graduate student. I wish also to recognize my indebtedness to Elda E. Anderson, Ph.D., and to William Zebrun, Ph.D. I shall not forget their encouragement during my early years of graduate study.

William R. Hendee

ACKNOWLEDGMENTS

One of the greatest pleasures of teaching is the continuing opportunity to work with former students on projects of mutual interest. This text is a good example of such an opportunity. Russ Ritenour received postdoctoral training in medical physics at the University of Colorado while I was chair of the Department of Radiology at that institution. After his NIH postdoctoral fellowship, he stayed on the faculty and we published several papers together, both before and after each of us left Colorado for new adventures. When it came time to write a 3rd edition of *Medical Imaging Physics* about ten years ago, I realized that I needed a co-author to share the workload. Russ was the person I wanted, and I am glad he agreed to be a co-author. He was equally willing to co-author this 4th edition. Future editions will bear his imprint as principal author of *Medical Imaging Physics*.

Several other persons deserve recognition for their support of this project. Foremost are Ms. Terri Komar and Ms. Mary Beth Drapp, both of whom have been instrumental in moving the fourth edition to completion. Terri worked with me as Executive Assistant for almost 10 years before moving to North Carolina. She was succeeded most ably in the position by Mary Beth Drapp. It has been my great fortune to be able to work in my publication efforts with two such competent individuals. Our editor, Ms. Luna Han of John Wiley Publishers, should be recognized for her quiet but firm insistence that we meet our own deadlines. I also am indebted to Jim Youker, M.D., Chair of Radiology at the Medical College of Wisconsin, for his friendship and inspiration over the years and for his enthusiasm for various academic ventures that we have collaborated in.

Most of all, I want to thank my wife Jeannie. Her tolerance of my writing habits, including stacks of books and papers perched on the piano, on the dining table, and, most precariously, in my study, is unfathomable. I certainly don't question it, but I do appreciate it—very much.

William R. Hendee, Ph.D.

I'm delighted to have been able to contribute once again to a new edition of this text and am particularly delighted to work once again with Bill Hendee. There was a lot to do, as so many things have changed and evolved in radiology since the time of the last edition in 1992. But, to me, the change is the fun part.

This was a fun project for another reason as well. Bill and I both enjoy using anecdotes as we teach. I'm referring to historical vignettes, illustrations of radiologic principles through examples in other fields, and, in my case I'm told, terrible jokes. While we felt that the terrible jokes were too informal for a textbook, we have included a number of vignettes and examples from other fields in the hope that it will make the reading more enjoyable and provide the kind of broader framework that leads to a deeper understanding. At least it might keep you awake.

I have to thank more people than there is space to thank. In particular, though, I must thank Pam Hansen, for dealing so patiently with many drafts of a complicated electronic manuscript. Also, I must thank two of my colleagues here at the University of Minnesota, Richard Geise and Bruce Hasselquist, who are endless sources of information and who never hesitate to tell me when I'm wrong. Rolph Gruetter of the Center for Magnetic Resonance Research, was very helpful in reviewing and commenting upon some of the new MR material in this edition. Finally, I want to thank Dr. William M. Thompson, who recently stepped down as chair of radiology here. He has been a tireless supporter of learning at all levels and he will be missed.

Once again, my wife, Julie, and our children, Jason and Karis, have supported me in so many ways during a major project. In this case, they've also been the source of a few of the medical images, although I won't say which ones.

E. Russell Ritenour, Ph.D.

CHAPTER

1

IMAGING IN MEDICINE

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OBJECTIVES

After completing this chapter, the reader should be able to:

- Identify the energy sources, tissue properties, and image properties employed in medical imaging.
- Name several factors influencing the increasing role of imaging in healthcare today.
- Define the expression "molecular medicine" and give examples.
- Provide a summary of the history of medical imaging.
- Explain the pivotal role of x-ray computed tomography in the evolution of modern medical imaging.

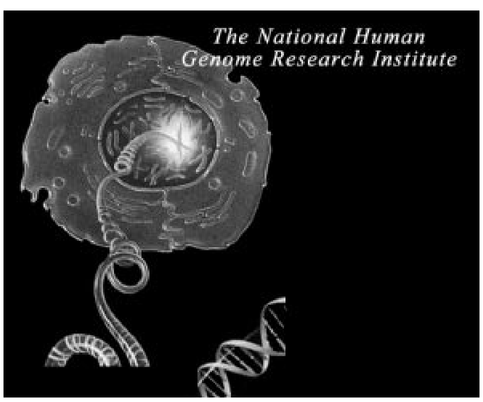
INTRODUCTION

Natural science is the search for "truth" about the natural world. In this definition, truth is defined by principles and laws that have evolved from observations and measurements about the natural world. The observations and measurements are reproducible through procedures that follow universal rules of scientific experimentation. They reveal properties of objects and processes in the natural world that are assumed to exist independently of the measurement technique and of our sensory perceptions of the natural world. The mission of science is to use observations and measurements to characterize the static and dynamic properties of objects, preferably in quantitative terms, and to integrate these properties into principles and, ultimately, laws and theories that provide a logical framework for understanding the world and our place in it.

Whether the external (natural) world can really be known, and even whether there is a world external to ourselves, has been the subject of philosophical speculation for centuries. It is for this reason that "truth" in the first sentence is offset in quotes.

It is not possible to characterize all properties of an object with exactness. For example, if the location of a particle is exactly known, its velocity is highly uncertain, and vice versa. Similarly, if the energy of a particle is exactly known, the time at which the particle has this energy is highly uncertain, and vice versa. This fundamental tenet of physics is known as the Heisenberg Uncertainty Principle.

As a part of natural science, human medicine is the quest for understanding one particular object, the human body, and its structure and function under all conditions of health, illness, and injury. This quest has yielded models of human health and illness that are immensely useful in preventing disease and disability, detecting and diagnosing illness and injury, and designing therapies to alleviate pain and suffering and to restore the body to a state of wellness or, at least, structural and functional capacity. The success of these efforts depends on (a) our depth of understanding of the human body and (b) the delineation of ways to intervene successfully in the progression of disease and the effects of injuries.



MARGIN FIGURE 1-1
The Human Genome project is a massive undertaking to determine the exact sequence of nucleotides (i.e., the DNA code) on all 24 human chromosomes.

Progress toward these objectives has been so remarkable that the average life span of humans in developed countries is almost twice its expected value a century ago. Greater understanding has occurred at all levels, from the atomic through molecular, cellular, and tissue to the whole body, and includes social and lifestyle influences on disease patterns. At present a massive research effort is focused on acquiring knowledge about genetic coding (the Human Genome Project) and about the role of genetic coding in human health and disease. This effort is progressing at an astounding rate, and it causes many medical scientists to believe that genetics, computational biology (mathematical modeling of biological systems), and bioinformatics (mathematical modeling of biological information, including genetic information) are the major research frontiers of medical science for the next decade or longer.

The number of deaths per 100,000 residents in the United States has declined from more than 400 in 1950 to less than 200 in 1990.

The human body is an incredibly complex system. Acquiring data about its static and dynamic properties results in massive amounts of information. One of the major challenges to researchers and clinicians is the question of how to acquire, process, and display vast quantities of information about the body so that the information can be assimilated, interpreted, and utilized to yield more useful diagnostic methods and therapeutic procedures. In many cases, the presentation of information as images is the most efficient approach to addressing this challenge. As humans we understand this efficiency; from our earliest years we rely more heavily on sight than on any other perceptual skill in relating to the world around us. Physicians increasingly rely

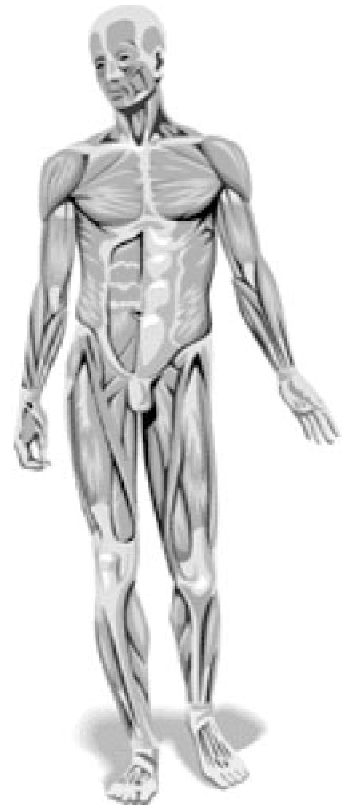
TABLE 1-1 Energy Sources and Tissue Properties Employed in Medical Imaging

Energy Sources	*Tissue Properties*	*Image Properties*
X rays	Mass density	Transmissivity
γ rays	Electron density	Opacity
Visible light	Proton density	Emissivity
Ultraviolet light	Atomic number	Reflectivity
Annihilation Radiation	Velocity	Conductivity
Electric fields	Pharmaceutical Location	Magnetizability
Magnetic fields	Current flow	Resonance
Infrared	Relaxation	Absorption
Ultrasound	Blood volume/flow	
Applied voltage	Oxygenation level of blood	
	Temperature	
	Chemical state	

as well on images to understand the human body and intervene in the processes of human illness and injury. The use of images to manage and interpret information about biological and medical processes is certain to continue its expansion, not only in clinical medicine but also in the biomedical research enterprise that supports it.

Images of a complex object such as the human body reveal characteristics of the object such as its transmissivity, opacity, emissivity, reflectivity, conductivity, and magnetizability, and changes in these characteristics with time. Images that reveal one or more of these characteristics can be analyzed to yield information about underlying properties of the object, as depicted in Table 1-1. For example, images (shadowgraphs) created by x rays transmitted through a region of the body reveal intrinsic properties of the region such as effective atomic number Z, physical density (grams/cm^3), and electron density (electrons/cm^3). Nuclear medicine images, including emission computed tomography (ECT) with pharmaceuticals releasing positrons [positron emission tomography (PET)] and single photons [single-photon emission computed tomography (SPECT)], reveal the spatial and temporal distribution of target-specific pharmaceuticals in the human body. Depending on the application, these data can be interpreted to yield information about physiological processes such as glucose metabolism, blood volume, flow and perfusion, tissue and organ uptake, receptor binding, and oxygen utilization. In ultrasonography, images are produced by capturing energy reflected from interfaces in the body that separate tissues with different acoustic impedances, where the acoustic impedance is the product of the physical density and the velocity of ultrasound in the tissue. Magnetic resonance imaging (MRI) of relaxation characteristics following magnetization of tissues is influenced by the concentration, mobility, and chemical bonding of hydrogen and, less frequently, other elements present in biological tissues. Maps of the electrical field (electroencephalography) and the magnetic field (magnetoencephalography) at the surface of the skull can be analyzed to identify areas of intense neuroelectrical activity in the brain. These and other techniques that use the energy sources listed in Table 1-1 provide an array of imaging methods that are immensely useful for displaying structural and functional information about the body. This information is essential to improving human health through detection and diagnosis of illness and injury.

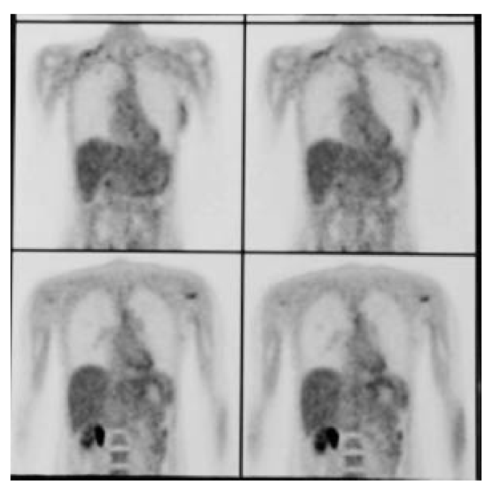
The intrinsic properties of biological tissues that are accessible through acquisition and interpretation of images vary spatially and temporally in response to structural and functional changes in the body. Analysis of these variations yields information about static and dynamic processes in the human body. These processes may be changed by disease and disability, and identification of the changes through imaging often permits detection and delineation of the disease or disability. Medical images are pictures of tissue characteristics that influence the way energy is emitted, transmitted, reflected, and so on, by the human body. These characteristics are related to, but not the same as, the actual structure (anatomy), composition (biology and



MARGIN FIGURE 1-2
Drawing of the human figure.

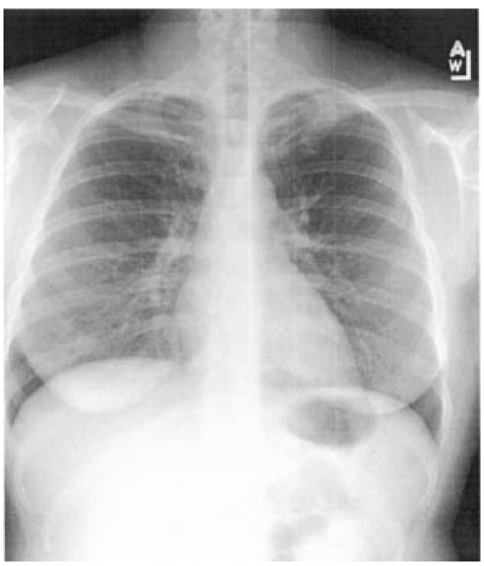
The effective atomic number Z_{eff} actually should be used in place of the atomic number Z in this paragraph. Z_{eff} is defined later in the text.

Promising imaging techniques that have not yet found applications in clinical medicine are discussed in the last chapter of the text.



MARGIN FIGURE 1-3
^{18}F-FDG PET scan of breast cancer patient with lymph node involvement in the left axilla.

MARGIN FIGURE 1-4
MRI of the human cervical spine.



MARGIN FIGURE 1-5
A normal chest radiograph. (Courtesy of Lacey Washington, M.D., Medical College of Wisconsin.)

Technology “push” means that technologies developed for specific applications, or perhaps for their own sake, are driven by financial incentives to find applications in other areas, including healthcare.

Sonar is an acronym for *SO*und *N*avigation *A*nd *R*anging.

The Manhattan Project was the code name for the U.S. project to develop a nuclear weapon during World War II.

chemistry), and function (physiology and metabolism) of the body. Part of the art of interpreting medical images is to bridge among image characteristics, tissue properties, human anatomy, biology and chemistry, and physiology and metabolism, as well as to determine how all of these parameters are affected by disease and disability.

Advances in Medical Imaging

Advances in medical imaging have been driven historically by the “technology push” principle. Especially influential have been imaging developments in other areas, notably in the defense and military sectors, that have been imported into medicine because of their potential applications to detection and diagnosis of human illness and injury. Examples include ultrasound developed initially for submarine detection (Sonar), scintillation detectors, and reactor-produced isotopes (including ^{131}I, ^{60}Co, and ^{99m}Tc) that emerged from the Manhattan Project, rare-earth fluorescent compounds synthesized initially in defense and space research laboratories, electrical devices for detection of rapid blood loss on the battlefield, and the evolution of microelectronics and computer industries from research funded initially for security, surveillance, defense, and military purposes. Basic research laboratories have also produced several imaging technologies that have migrated successfully into clinical medicine. Examples include (a) reconstruction mathematics for computed tomographic imaging and (b) laboratory techniques in nuclear magnetic resonance that evolved into magnetic resonance imaging, spectroscopy, and other methods useful in clinical medicine. The migration of technologies from other arenas into medicine has not always been successful. For example, infrared detection devices developed for night vision in military operations have so far not proven to be useful in medicine in spite of initial enthusiasm for infrared thermography as an imaging method for early detection of breast cancer.

Today the emphasis in medical imaging is shifting from a “technology push” approach toward a “biological/clinical pull” emphasis. This shift reflects both (a) a deeper understanding of the biology underlying human health and disease and (b) a growing demand for accountability (proven usefulness) of technologies before they are introduced into clinical medicine. Increasingly, unresolved biological questions important to the diagnosis and treatment of human disease and disability are used to encourage development of new imaging methods, often in association with nonimaging probes. For example, the functions of the human brain, along with the causes and mechanisms of various mental disorders such as dementia, depression, and schizophrenia, are among the greatest biological enigmas confronting biomedical scientists and clinicians. A particularly fruitful method for penetrating this conundrum is the technique of functional imaging employing tools such as ECT and MRI. Functional magnetic resonance imaging (fMRI) is especially promising as an approach to unraveling some of the mysteries related to how the human brain functions in health, disease, and disability. Another example is the use of x-ray computed tomography and MRI as feedback mechanisms to shape, guide, and monitor the surgical and radiation treatment of cancer.

The growing use of imaging techniques in radiation oncology reveals an interesting and rather recent development. Until about three decades ago, the diagnostic and therapeutic applications of ionizing radiation were practiced by a single medical specialty. In the late 1960s these applications began to separate into distinct medical specialties, diagnostic radiology and radiation oncology, with separate training programs and clinical practices. Today, imaging is used extensively in radiation oncology to characterize the cancers to be treated, design the plans of treatment, guide the delivery of radiation, monitor the response of patients to treatment, and follow patients over the long term to assess the success of therapy, occurrence of complications, and frequency of recurrence. The process of accommodating to this development in the training and practice of radiation oncology is encouraging a closer working relationship between radiation oncologists and diagnostic radiologists.

Evolutionary Developments in Imaging

Six major developments are converging today to raise imaging to a more prominent role in biological and medical research and in the clinical practice of medicine. These developments are[1]:

- Ever-increasing sophistication of the biological questions that can be addressed as knowledge expands and understanding grows about the complexity of the human body and its static and dynamic properties.
- Ongoing evolution of imaging technologies and the increasing breadth and depth of the questions that these technologies can address at ever more fundamental levels.
- Accelerating advances in computer technology and information networking that support imaging advances such as three- and four-dimensional representations, superposition of images from different devices, creation of virtual reality environments, and transportation of images to remote sites in real time.
- Growth of massive amounts of information about patients that can best be compressed and expressed through the use of images.
- Entry into research and clinical medicine of young persons who are highly facile with computer technologies and comfortable with images as the principal pathway to information acquisition and display.
- Growing importance of images as effective means to convey information in visually-oriented developed cultures.

Medical Imaging Trends

From	To
Anatomic	Physiobiochemical
Static	Dynamic
Qualitative	Quantitative
Analog	Digital
Nonspecific agents	Tissue-Targeted agents
Diagnosis	Diagnosis/Therapy

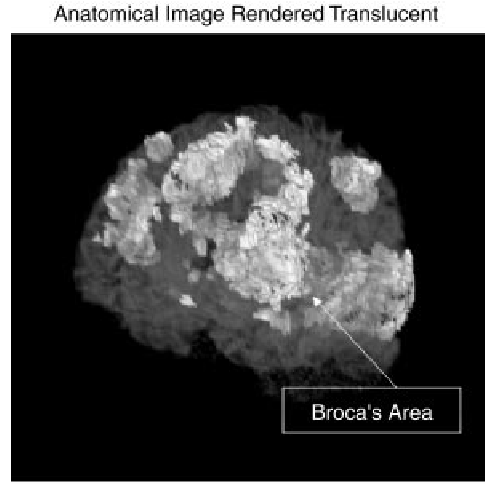
Biological/clinical "pull" means that technologies are developed in response to recognized clinical or research needs.

A major challenge confronting medical imaging today is the need to efficiently exploit this convergence of evolutionary developments to accelerate biological and medical imaging toward the realization of its true potential.

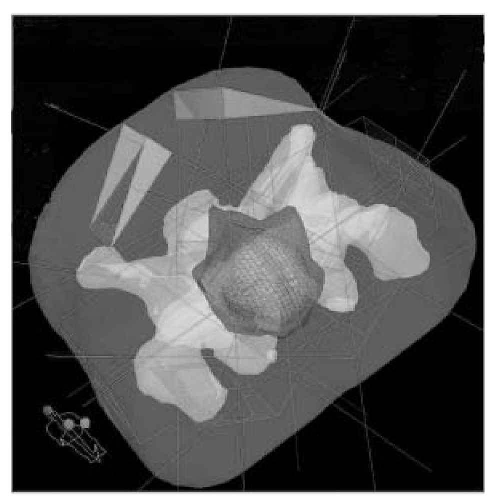
Images are our principal sensory pathway to knowledge about the natural world. To convey this knowledge to others, we rely on verbal communication following accepted rules of human language, of which there are thousands of varieties and dialects. In the distant past, the acts of knowing through images and communicating through languages were separate and distinct processes. Every technological advance that brought images and words closer, even to the point of convergence in a single medium, has had a major cultural and educational impact. Examples of such advances include the printing press, photography, motion pictures, television, video games, computers, and information networking. Each of these technologies has enhanced the shift from using words to communicate information toward a more efficient synthesis of images to provide insights and words to explain and enrich insights.[2] Today this synthesis is evolving at a faster rate than ever before, as evidenced, for example, by the popularity of television news programs and documentaries and the growing use of multimedia approaches to education and training.

For purposes of informing and educating individuals, multiple pathways are required for interchanging information. In addition, flexible means are needed for mixing images and words, and their rate and sequence of presentation, in order to capture and retain the attention, interest, and motivation of persons engaged in the educational process. Computers and information networks provide this capability. In medicine, their use in association with imaging technologies greatly enhances the potential contribution of medical imaging to resolution of patient problems in the clinical setting. At the beginning of the twenty-first century, the six evolutionary developments discussed above provide the framework for major advances in medical imaging and its contributions to improvements in the health and well-being of people worldwide.

Anatomical Image Rendered Translucent



MARGIN FIGURE 1-6
fMRI image of brain tumor in relation to the motor cortex.



MARGIN FIGURE 1-7
Multifield treatment plan superimposed on a 3-D reconstructed CT image. (From G. Ibbott, Ph.D., MD Anderson Hospital. Used with permission.)

Molecular Medicine

Medical imaging has traditionally focused on the acquisition of structural (anatomic) and functional (physiologic) information about patients at the organ and tissue levels.

If a scientist reads two articles each day from the world's scientific literature published that day, at the end of one year the scientist will be 60 centuries behind in keeping up with the current scientific literature. To keep current with the literature, the scientist would have to read 6000 articles each day.

Each new generation adapts with ease to technologies that were a challenge to the previous generation. Examples of this "generation gap" in today's world include computers, software engineering, and video games.

Imaging technologies useful or potentially useful at the cellular and molecular levels:

- Multiphoton microscopy
- Scanning probe microscopy
- Electron energy-loss spectroscopic imaging
- Transmission electron microscopes with field-emission electron guns
- 3-D reconstruction from electron micrographs
- Fluorescent labels
- Physiologic indicators
- Magnetic resonance imaging microscopy
- Single-copy studies of proteins and oligonucleotides
- Video microscopy
- Laser-scanning confocal microscopy
- Two-photon laser scanning microscopy

Antisense agents (molecules, viruses, etc.) are agents that contain DNA with a nucleotide configuration opposite that of the biological structures for which the agents are targeted.

A major challenge to the use of molecular mechanisms to enhance contrast are limitations on the number of cells that can be altered by various approaches.

This focus has nurtured the correlation of imaging findings with pathological conditions and has led to substantial advances in detection and diagnosis of human disease and injury. All too often, however, detection and diagnosis occur at a stage in the disease or injury where radical intervention is required and the effectiveness of treatment is compromised. In many of these cases, detection and diagnosis at an earlier stage in the progression of disease and injury would improve the effectiveness of treatment and enhance the well-being of patients. This objective demands that medical imaging expand its focus from the organ and tissue levels to the cellular and molecular levels of human disease and injury. Many scientists believe that medical imaging is well-positioned today to experience this expanded focus as a benefit of knowledge gained at the research frontiers of molecular biology and genetics. This benefit is often characterized as the entry of medical imaging into the era of molecular medicine.

Contrast agents are widely employed with x-ray, ultrasound, and magnetic resonance imaging techniques to enhance the visualization of properties correlated with patient anatomy and physiology. Agents in wide use today localize in tissues either by administration into specific anatomic compartments (such as the gastrointestinal or vascular systems) or by reliance on nonspecific changes in tissues (such as increased capillary permeability or alterations in the extracellular fluid space). These localization mechanisms frequently do not provide a sufficient concentration of the agent to reveal subtle tissue differences associated with an abnormal condition. New contrast agents are needed that exploit growing knowledge about biochemical receptor systems, metabolic pathways, and "antisense" molecular technologies to yield concentration differentials sufficient to reveal the presence of pathological conditions.

Another important imaging application of molecular medicine is the use of imaging methods to study molecular and genetic processes. For example, cells may be genetically altered to attract ions that (1) alter the magnetic susceptibility, thereby permitting their identification by magnetic resonance imaging techniques; or (2) are radioactive and therefore can be visualized by nuclear imaging methods. Another possibility is to transect cells with genetic material that causes expression of cell surface receptors that can bind radioactive compounds.[3] Conceivably, this technique could be used to monitor the progress of gene therapy.

Advances in molecular biology and genetics are yielding new knowledge at an astonishing rate about the molecular and genetic infrastructure underlying the static and dynamic processes that comprise human anatomy and physiology. This new knowledge is likely to yield increasingly specific approaches to the use of imaging methods to visualize normal and abnormal tissue structure and function at increasingly fundamental levels. These methods will in all likelihood contribute to continuing advances in molecular medicine.

Historical Approaches to Diagnosis

In the 1800s and before, physicians were extremely limited in their ability to obtain information about the illnesses and injuries of patients. They relied essentially on the five human senses, and what they could not see, hear, feel, smell, or taste usually went undetected. Even these senses could not be exploited fully, because patient modesty and the need to control infectious diseases often prevented full examination of the patient. Frequently, physicians served more to reassure the patient and comfort the family rather than to intercede in the progression of illness or facilitate recovery from injury. More often than not, fate was more instrumental than the physician in determining the course of a disease or injury.

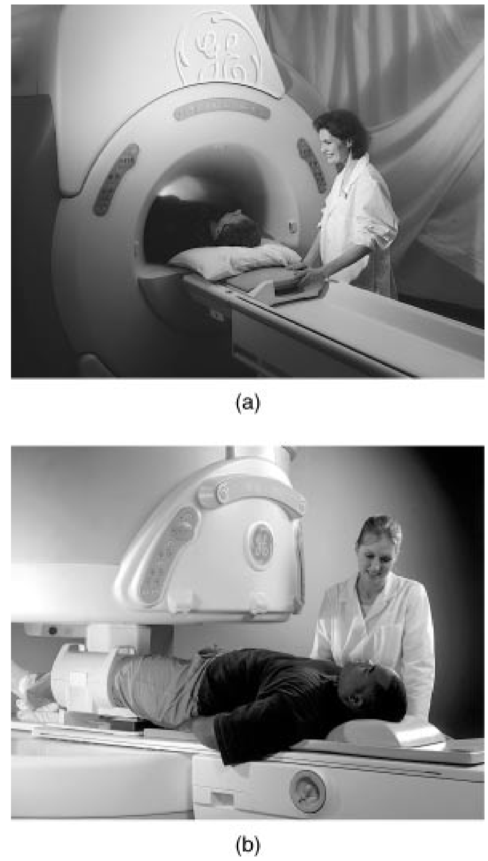
The twentieth century witnessed remarkable changes in the physician's ability to intervene actively on behalf of the patient. These changes dramatically improved the health of humankind around the world. In developed countries, infant mortality decreased substantially, and the average life span increased from 40 years at the beginning of the century to 70+ years at the century's end. Many major diseases, such as smallpox, tuberculosis, poliomyelitis, and pertussis, had been brought under control, and some had been virtually eliminated. Diagnostic medicine has improved dramatically, and therapies have evolved for cure or maintenance of persons with a variety of maladies.

Diagnostic probes to identify and characterize problems in the internal anatomy and physiology of patients have been a major contribution to these improvements. By far, x rays are the most significant of these diagnostic probes. Diagnostic x-ray studies have been instrumental in moving the physician into the role of an active intervener in disease and injury and a major influence on the prognosis for recovery.

Capsule History of Medical Imaging

In November 1895 Wilhelm Röntgen, a physicist at the University of Würzburg, was experimenting with cathode rays. These rays were obtained by applying a potential difference across a partially evacuated glass "discharge" tube. Röntgen observed the emission of light from crystals of barium platinocyanide some distance away, and he recognized that the fluorescence had to be caused by radiation produced by his experiments. He called the radiation "x rays" and quickly discovered that the new radiation could penetrate various materials and could be recorded on photographic plates. Among the more dramatic illustrations of these properties was a radiograph of a hand (Figure 1-1) that Röntgen included in early presentations of his findings.[4] This radiograph captured the imagination of both scientists and the public around the world.[5] Within a month of their discovery, x rays were being explored as medical tools in several countries, including Germany, England, France, and the United States.[6]

Two months after Röntgen's discovery, Poincaré demonstrated to the French Academy of Sciences that x rays were released when cathode rays struck the wall of a gas discharge tube. Shortly thereafter, Becquerel discovered that potassium uranyl sulfate spontaneously emitted a type of radiation that he termed Becquerel rays, now



MARGIN FIGURE 1-8
Closed-bore (**top**) and open-bore (**bottom**) MRI units. (Courtesy of General Electric Medical Systems.)

FIGURE 1-1
A radiograph of the hand taken by Röntgen in December 1895. His wife may have been the subject. (From the Deutsches Röntgen Museum, Remscheid-Lennap, Germany. Used with permission.)

In 1901, Röntgen was awarded the first Nobel Prize in Physics.

popularly known as β-particles.[7] Marie Curie explored Becquerel rays for her doctoral thesis and chemically separated a number of elements. She discovered the radioactive properties of naturally occurring thorium, radium, and polonium, all of which emit α-particles, a new type of radiation.[8] In 1900, γ rays were identified by Villard as a third form of radiation.[9] In the meantime, J. J. Thomson reported in 1897 that the cathode rays used to produce x rays were negatively charged particles (electrons) with about 1/2000 the mass of the hydrogen atom.[10] In a period of 5 years from the discovery of x rays, electrons and natural radioactivity had also been identified, and several sources and properties of the latter had been characterized.

Over the first half of the twentieth century, x-ray imaging advanced with the help of improvements such as intensifying screens, hot-cathode x-ray tubes, rotating anodes, image intensifiers, and contrast agents. These improvements are discussed in subsequent chapters. In addition, x-ray imaging was joined by other imaging techniques that employed radioactive nuclides and ultrasound beams as radiation sources for imaging.

Through the 1950s and 1960s, diagnostic imaging progressed as a coalescence of x-ray imaging with the emerging specialties of nuclear medicine and ultrasonography. This coalescence reflected the intellectual creativity nurtured by the synthesis of basic science, principally physics, with clinical medicine. In a few institutions, the interpretation of clinical images continued to be taught without close attention to its foundation in basic science. In the more progressive teaching departments, however, the dependence of radiology on basic science, especially physics, was never far from the consciousness of teachers and students.

In the early years of CT, an often-heard remark was "why would anyone want a new x-ray technique that when compared with traditional x-ray imaging:

- yields 10 times more coarse spatial resolution
- is 1/100 as fast in collecting image data
- costs 10 times more

Introduction of Computed Tomography

In the early 1970s a major innovation was introduced into diagnostic imaging. This innovation, x-ray computed tomography (CT), is recognized today as the most significant single event in medical imaging since the discovery of x rays.

The importance of CT is related to several of its features, including the following:

1. Provision of cross-sectional images of anatomy
2. Availability of contrast resolution superior to traditional radiology
3. Construction of images from x-ray transmission data by a "black box" mathematical process requiring a computer
4. Creation of clinical images that are no longer direct proof of a satisfactory imaging process so that intermediate control measures from physics and engineering are essential
5. Production of images from digital data that are processed by computer and can be manipulated to yield widely varying appearances.

EMI Ltd., the commercial developer of CT, was the first company to enter CT into the market. They did so as a last resort, only after offering the rights to sell, distribute, and service CT to the major vendors of imaging equipment. The vendors rejected EMI's offer because they believed the market for CT was too small.

Adoption of CT by the medical community was rapid and enthusiastic in the United States and worldwide. A few years after introduction of this technology, more than 350 units had been purchased in the United States alone. Today, CT is an essential feature of most radiology departments of moderate size and larger.

The introduction of CT marked the beginning of a transition in radiology from an analog to a digitally based specialty. The digital revolution in radiology has opened opportunities for image manipulation, storage, transmission, and display in all fields of medicine. The usefulness of CT for brain imaging almost immediately reduced the need for nuclear brain scans and stimulated the development of other applications of nuclear medicine, including qualitative and quantitative studies of the cardiovascular system. Extension of reconstruction mathematics to nuclear medicine yielded the techniques of single-photon emission computed tomography (SPECT) and positron emission tomography (PET), technologies that have considerable potential for revealing new information about tissue physiology and metabolism. Reconstruction mathematics also are utilized in magnetic resonance image (MRI), a technology introduced into clinical medicine in the early 1980s. Today, MRI provides insights into fundamental properties of biologic tissues that were beyond the imagination a few years ago. Digital methods have been incorporated into ultrasonography to provide "real time" gray scale images important to the care of patients in cardiology, obstetrics, and several other specialties. In x-ray imaging, digital methods are slowly but inexorably replacing analog methods for data acquisition and display.

Many believe that the next major frontier of imaging science is at the molecular and genetic levels.

Radiology is a much different field today than it was three decades ago. With the introduction of new imaging methods and digital processing techniques, radiology has become a technologically complex discipline that presents a paradox for physicians. Although images today are much more complicated to produce, they are simultaneously simpler to interpret—and misinterpret—once they are produced. The simplicity of image interpretation is seductive, however. The key to retrieval of essential information in radiology today resides at least as much in the production and presentation of images as in their interpretation.

A physician who can interpret only what is presented as an image suffers a severe handicap. He or she is captive to the talents and labors of others and wholly dependent on their ability to ensure that an image reveals abnormalities in the patient and not in the imaging process. On the other hand, the physician who understands the science and technology of imaging can be integrally involved in the entire imaging process, including the acquisition of patient data and their display as clinical images. Most important, the knowledgeable physician has direct input into the quality of the image on which the diagnosis depends.

A thorough understanding of diagnostic images requires knowledge of the science, principally physics, that underlies the production of images. Radiology and physics have been closely intertwined since x rays were discovered. With the changes that have occurred in imaging over the past few years, the linkage between radiology and physics has grown even stronger. Today a reasonable knowledge of physics, instrumentation, and imaging technology is essential for any physician wishing to perfect the science and art of radiology.

It is wrong to think that the task of physics is to find out what nature is. Physics concerns what we can say about nature." Niels Bohr (as quoted in Pagels, H., *The Cosmic Code*, Simons and Schuster, 1982.)

CONCLUSIONS

- Medical imaging is both a science and a tool to explore human anatomy and to study physiology and biochemistry.
- Medical imaging employs a variety of energy sources and tissue properties to produce useful images.
- Increasingly, clinical pull is the driving force in the development of imaging methods.
- Molecular biology and genetics are new frontiers for imaging technologies.
- Introduction of x-ray computed tomography was a signal event in the evolution of medical imaging.

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CHAPTER

2

STRUCTURE OF MATTER

Medical Imaging Physics, Fourth Edition, by William R. Hendee and E. Russell Ritenour
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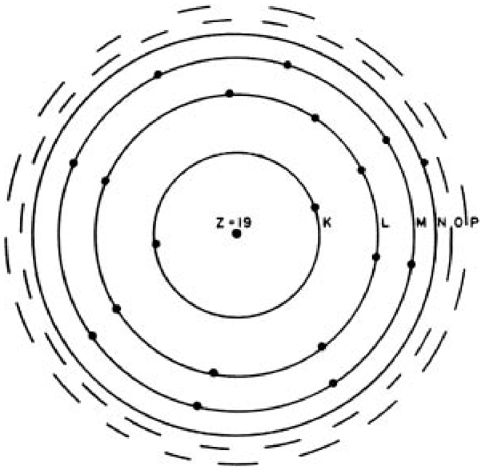


MARGIN FIGURE 2-1
Electron configuration showing electron shells in the Bohr model of the atom for potassium, with 19 electrons ($Z = 19$).

A Brief History of the Development of Evidence for the Existence of Atoms
400–300 B.C.: Demokritos and the Epicurean School in Greece argued for the existence of atoms on philosophical grounds. Aristotle and the Stoics espoused the continuum philosophy of matter.[1]
300 B.C.–1800s: Aristotelian view of matter predominated.
1802: Dalton described the principle of multiple proportions. This principle states that chemical constituents react in specific proportions, suggesting the discreteness of material components.[2]
1809: Gay-Lussac discovered laws that predicted changes in volume of gases.[3]
1811: Avogadro hypothesized the existence of a constant number of atoms in a characteristic mass of an element or compound.[4]
1833: Faraday's law of electrolysis explained specific rates or proportions of elements that would be electroplated onto electrodes from electrolytic solutions.[5]
1858: Cannizaro published data concerning the atomic weights of the elements.[6]
1869–1870: Meyer and Mendeleev constructed the Periodic Table.[7, 8]
1908: Perrin demonstrated that the transfer of energy from atoms to small particles in solution, the cause of a phenomenon known as Brownian motion, leads to a precise derivation of Avogadro's number.[9]

OBJECTIVES

After completing this chapter, the reader should be able to:

- Define the terms: element, atom, molecule, and compound.
- Describe the electron shell structure of an atom.
- Explain the significance of electron and nuclear binding energy.
- List the events that result in characteristic and auger emission.
- Compare electron energy levels in solids that are:
  - Conductors
  - Insulators
  - Semiconductors
- Describe the phenomenon of superconductivity.
- List the four fundamental forces.
- Explain why fission and fusion result in release of energy.
- State the source of the nuclear magnetic moment.
- Define:
  - Isotopes
  - Isotones
  - Isobars
  - Isomers

THE ATOM

All matter is composed of atoms. A sample of a pure element is composed of a single type of atom. Chemical compounds are composed of more than one type of atom. Atoms themselves are complicated entities with a great deal of internal structure. An atom is the smallest unit of matter that retains the chemical properties of a material. In that sense, it is a "fundamental building block" of matter. In the case of a compound, the "fundamental building block" is a molecule consisting of one or more atoms bound together by electrostatic attraction and/or the sharing of electrons by more than one nucleus.

The basic structure of an atom is a positively charged nucleus, containing electrically neutral neutrons and positively charged protons, surrounded by one or more negatively charged electrons. The number and distribution of electrons in the atom determines the chemical properties of the atom. The number and configuration of neutrons and protons in the nucleus determines the stability of the atom and its electron configuration.

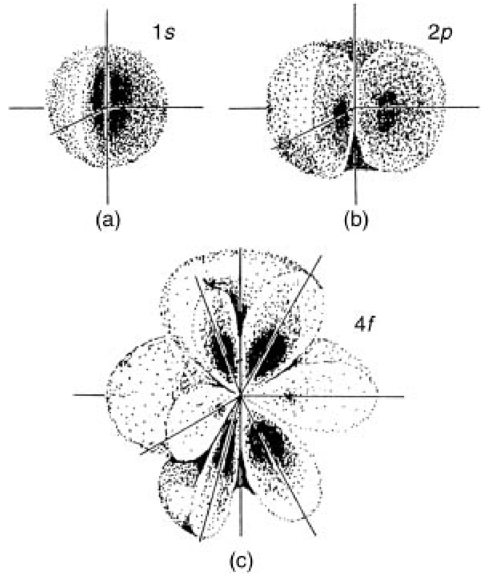
Structure of the Atom

One unit of charge is 1.6×10^{-19} coulombs. Each proton and each electron carries one unit of charge, with protons positive and electrons negative. The number of units of positive charge (i.e., the number of protons) in the nucleus is termed the atomic number Z. The atomic number uniquely determines the classification of an atom as one of the elements. Atomic number 1 is hydrogen, 2 is helium, and so on.

Atoms in their normal state are neutral because the number of electrons outside the nucleus (i.e., the negative charge in the atom) equals the number of protons (i.e., the positive charge) of the nucleus. Electrons are positioned in energy levels (i.e., shells) that surround the nucleus. The first ($n = 1$) or K shell contains no more than 2 electrons, the second ($n = 2$) or L shell contains no more than 8 electrons, and the third ($n = 3$) or M shell contains no more than 18 electrons (Margin Figure 2-1). The outermost electron shell of an atom, no matter which shell it is, never contains more than 8 electrons. Electrons in the outermost shell are termed valence electrons and determine to a large degree the chemical properties of the atom. Atoms with an outer shell entirely filled with electrons seldom react chemically. These atoms constitute elements known as the inert gases (helium, neon, argon, krypton, xenon, and radon).

TABLE 2-1 Quantum Numbers for Electrons in Helium, Carbon, and Sodium

Element	*n*	*l*	m_l	m_s	*Orbital*	*Shell*
Helium($Z = 2$)	1	0	0	$-1/2$	$1s^2$	K
	1	0	0	$+1/2$		
Carbon($Z = 6$)	1	0	0	$-1/2$	$1s^2$	
	1	0	0	$+1/2$		
	2	0	0	$-1/2$	$2s^2$	L
	2	0	0	$+1/2$		
	2	1	−1	$-1/2$	$2p^2$	
	2	1	−1	$+1/2$		
Sodium($Z = 11$)	1	0	0	$-1/2$	$1s^2$	K
	1	0	0	$+1/2$		
	2	0	0	$-1/2$	$2s^2$	L
	2	0	0	$+1/2$		
	2	1	−1	$-1/2$	$2p^6$	
	2	1	−1	$+1/2$		
	2	1	0	$-1/2$		
	2	1	0	$+1/2$		
	2	1	1	$-1/2$		
	2	1	1	$+1/2$		
	3	0	0	$-1/2$	$3s$	M



MARGIN FIGURE 2-2
Probability of location of the electron in the hydrogen atom for three different energy states or combinations of principal and angular momentum quantum numbers l. **A**: $n = 1, l = 0$; **B**: $n = 2, l = 1$; **C**: $n = 4, l = 3$.

Energy levels for electrons are divided into sublevels slightly separated from each other. To describe the position of an electron in the extranuclear structure of an atom, the electron is assigned four quantum numbers.

The principal quantum number n defines the main energy level or shell within which the electron resides ($n = 1$ for the K shell, $n = 2$ for the L shell, etc.). The orbital angular-momentum (azimuthal) quantum number l describes the electron's angular momentum ($l = 0, 1, 2, \ldots, n - 1$). The orientation of the electron's magnetic moment in a magnetic field is defined by the magnetic quantum number m_l ($m_l = -l$, $-l + 1, \ldots, l - 1, l$). The direction of the electron's spin upon its own axis is specified by the spin quantum number m_s ($m_s = +1/2$ or $-1/2$). The Pauli exclusion principle states that no two electrons in the same atomic system may be assigned identical values for all four quantum numbers. Illustrated in Table 2-1 are quantum numbers for electrons in a few atoms with low atomic numbers.

The values of the orbital angular-momentum quantum number, $l = 0$, 1, 2, 3, 4, 5 and 6, are also identified with the symbols, *s, p, d, f, g, h,* and i, respectively. This notation is known as "spectroscopic" notation because it is used to describe the separate emission lines observed when light emitted from a heated metallic vapor lamp is passed through a prism. From the 1890s onward, observation of these spectra provided major clues about the binding energies of electrons in atoms of the metals under study. By the 1920s, it was known that the spectral lines above s could be split into multiple lines in the presence of a magnetic field. The lines were thought to correspond to "orbitals" or groupings of similar electrons within orbits. The modern view of this phenomenon is that, while the s "orbital" is spherically symmetric (Margin Figure 2-2), the other orbitals are not. In the presence of a magnetic field, the p "orbital" can be in alignment along any one of three axis of space, x, y, and z. Each of these three orientations has a slightly different energy corresponding to the three possible values of m_l (−1, 0, and 1). According to the Pauli exclusion principle, each orbital may contain two electrons (one with $m_s = +1/2$, the other with $m_s = -1/2$).

The K shell of an atom consists of one orbital, the $1s$, containing two electrons. The L shell consists of the $2s$ subshell, which contains one orbital (two electrons), and the $2p$ subshell, which contains a maximum of three orbitals (six electrons). If an L shell of an atom is filled, its electrons will be noted in spectroscopic notation as $2s^2$, $2p^6$. This notation is summarized for three atoms—helium, carbon, and sodium—in Table 2-1.

Models of the Atom
1907: J. J. Thompson advanced a "plum pudding" model of the atom in which electrons were distributed randomly within a matrix of positive charge, somewhat like raisins in a plum pudding.[10]
1911: Experiments by Rutherford showed the existence of a small, relatively dense core of positive charge in the atom, surrounded by mostly empty space with a relatively small population of electrons.[11]

Why are electron shells identified as K, L, M, and so on, instead of A,B,C, and so on?
Between 1905 and 1911, English physicist Charles Barkla measured the characteristic emission of x rays from metals in an attempt to categorize them according to their penetrating power (energy). Early in his work, he found two and named them B and A. In later years, he renamed them K and L, expecting that he would find more energetic emissions to name B, A, and R. It is thought that he was attempting to use letters from his name. Instead, he discovered lower energy (less penetrating) emissions and decided to continue the alphabet after L with M and N. The naming convention was quickly adopted by other researchers. Thus electron shells (from which characteristic x rays are emitted) are identified as K, L, M, N, and so on. Elements with shell designations up to the letter Q have been identified.[12]

Quantum Mechanical Description of Electrons

Since the late 1920s it has been understood that electrons in an atom do not behave exactly like tiny moons orbiting a planet-like nucleus. Their behavior is described more accurately if, instead of defining them as point particles in orbits with specific velocities and positions, they are defined as entities whose behavior is described by "wave functions." While a wave function itself is not directly observable, calculations may be performed with this function to predict the location of the electron. In contrast to the calculations of "classical mechanics" in which properties such as force, mass, acceleration, and so on, are entered into equations to yield a definite answer for a quantity such as position in space, quantum mechanical calculations yield probabilities. At a particular location in space, for example, the square of the amplitude of a particle's wave function yields the probability that the particle will appear at that location. In Margin Figure 2-2, this probability is predicted for several possible energy levels of a single electron surrounding a hydrogen nucleus (a single proton). In this illustration, a darker shading implies a higher probability of finding the electron at that location. Locations at which the probability is maximum correspond roughly to the "electron shell" model discussed previously. However, it is important to emphasize that the probability of finding the electron at other locations, even in the middle of the nucleus, is not zero. This particular result explains a certain form of radioactive decay in which a nucleus "captures" an electron. This event is not explainable by classical mechanics, but can be explained with quantum mechanics.

Electron Binding Energy and Energy Levels

The extent to which electrons are bound to the nucleus determines several energy absorption and emission phenomena. The binding energy of an electron (E_b) is defined as the energy required to completely separate the electron from the atom. When energy is measured in the macroscopic world of everyday experience, units such as joules and kilowatt-hours are used. In the microscopic world, the electron volt is a more convenient unit of energy. In Margin Figure 2-3 an electron is accelerated between two electrodes. That is, the electron is repelled from the negative electrode and attracted to the positive electrode. The kinetic energy (the "energy of motion") of the electron depends on the potential difference between the electrodes. One electron volt is the kinetic energy imparted to an electron accelerated across a potential difference (i.e., voltage) of 1 V. In Margin Figure 2-3B, each electron achieves a kinetic energy of 10 eV.

The electron volt may be used to express any level of energy. For example, a typical burner on "high" on an electric stove emits heat at an approximate rate of 3×10^{22} eV/sec.

The electron volt can be converted to other units of energy:

$$\begin{aligned} 1\text{ eV} &= 1.6 \times 10^{-19}\text{ J} \\ &= 1.6 \times 10^{-12}\text{ erg} \\ &= 4.4 \times 10^{-26}\text{ kW-hr} \end{aligned}$$

$$\textit{Note}: 10^3\text{ eV} = 1\text{ keV}$$

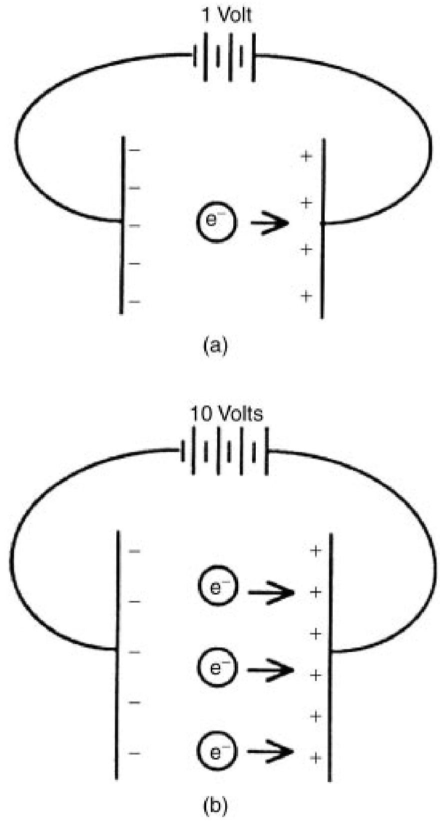
$$10^6\text{ eV} = 1\text{ MeV}$$

The electron volt describes potential as well as kinetic energy. The binding energy of an electron in an atom is a form of potential energy.

An electron in an inner shell of an atom is attracted to the nucleus by a force greater than that exerted by the nucleus on an electron farther away. An electron may be moved from one shell to another shell that is farther from the nucleus only if energy is supplied by an external source. Binding energy is negative (i.e., written with a minus sign) because it represents an amount of energy that must be supplied to remove an electron from an atom. The energy that must be imparted to an atom to move an electron from an inner shell to an outer shell is equal to the arithmetic difference in binding energy between the two shells. For example, the binding energy is −13.5 eV for an electron in the K shell of hydrogen and is −3.4 eV for an electron in the L shell. The energy required to move an electron from the K to the L shell in hydrogen is (−3.4 eV) − (−13.5 eV) = 10.1 eV.

Electrons in inner shells of high-Z atoms are near a nucleus with high positive charge. These electrons are bound to the nucleus with a force much greater than that exerted upon the solitary electron in hydrogen. Binding energies of electrons in high- and low-Z atoms are compared in Margin Figure 2-4.

All of the electrons within a particular electron shell do not have exactly the same binding energy. Differences in binding energy among the electrons in a particular shell are described by the orbital, magnetic, and spin quantum numbers, l, m_l, and m_s. The combinations of these quantum numbers allowed by quantum mechanics provide three subshells (L_I to L_{III}) for the L shell (Table 2-1) and five subshells (M_I to M_V) for the M shell (the M subshells occur only if a magnetic field is present). Energy differences between the subshells are small when compared with differences between shells. These differences are important in radiology, however, because they explain certain properties of the emission spectra of x-ray tubes. Table 2-2 gives values for the binding energies of K, and L shell electrons for selected elements.



MARGIN FIGURE 2-3
Kinetic energy of electrons specified in electron volts. **A**: The electron has a kinetic energy of 1 eV. **B**: Each electron has a kinetic energy of 10 eV.

Electron Transitions, Characteristic and Auger Emission

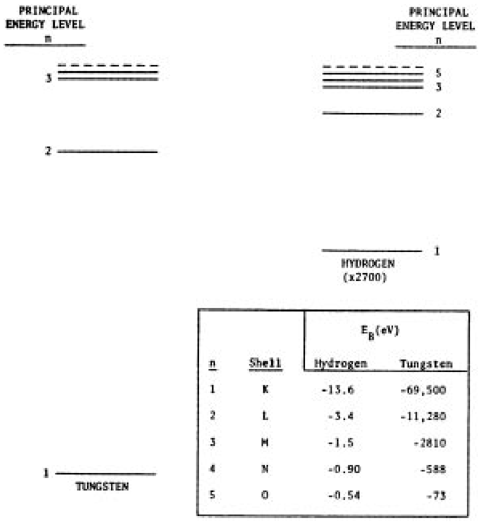
Various processes can cause an electron to be ejected from an electron shell. When an electron is removed from a shell, a vacancy or "hole" is left in the shell (i.e., a quantum "address" is left vacant). An electron may move from one shell to another to fill the vacancy. This movement, termed an *electron transition*, involves a

TABLE 2-2 Binding Energies of Electron Shells of Selected Elements[a]

		Binding Energies of Shells (keV)			
Atomic Number	*Element*	*K*	*L_I*	*L_{II}*	*L_{III}*
1	Hydrogen	0.0136			
6	Carbon	0.283			
8	Oxygen	0.531			
11	Sodium	1.08	0.055	0.034	0.034
13	Aluminum	1.559	0.087	0.073	0.072
14	Silicon	1.838	0.118	0.099	0.098
19	Potassium	3.607	0.341	0.297	0.294
20	Calcium	4.038	0.399	0.352	0.349
26	Iron	7.111	0.849	0.721	0.708
29	Copper	8.980	1.100	0.953	0.933
31	Gallium	10.368	1.30	1.134	1.117
32	Germanium	11.103	1.42	1.248	1.217
39	Yttrium	17.037	2.369	2.154	2.079
42	Molybdenum	20.002	2.884	2.627	2.523
47	Silver	25.517	3.810	3.528	3.352
53	Iodine	33.164	5.190	4.856	4.559
54	Xenon	34.570	5.452	5.104	4.782
56	Barium	37.410	5.995	5.623	5.247
57	Lanthanum	38.931	6.283	5.894	5.489
58	Cerium	40.449	6.561	6.165	5.729
74	Tungsten	69.508	12.090	11.535	10.198
79	Gold	80.713	14.353	13.733	11.919
82	Lead	88.001	15.870	15.207	13.044
92	Uranium	115.591	21.753	20.943	17.163

[a] Data from Fine, J., and Hendee, W. *Nucleonics* 1955; **13**:56.

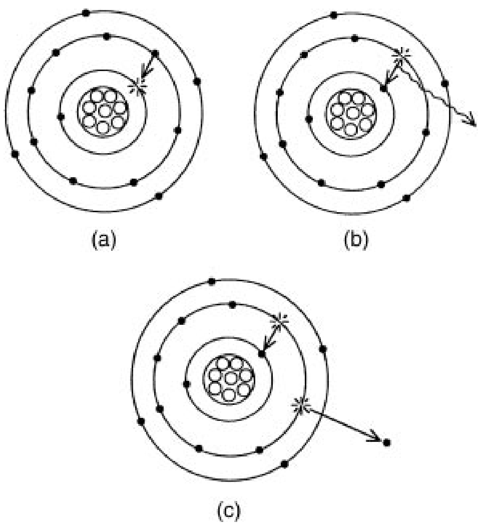
"No rest is granted to the atoms throughout the profound void, but rather driven by incessant and varied motions, some after being pressed together then leap back with wide intervals, some again after the blow are tossed about within a narrow compass. And those being held in combination more closely condensed collide with and leap back through tiny intervals."
Lucretius (94–55 B.C.), *On the Nature of Things*



		E_B (eV)	
n	Shell	Hydrogen	Tungsten
1	K	-13.6	-69,500
2	L	-3.4	-11,280
3	M	-1.5	-2810
4	N	-0.90	-588
5	O	-0.54	-73

MARGIN FIGURE 2-4
Average binding energies for electrons in hydrogen ($Z = 1$) and tungsten ($Z = 74$). Note the change of scale required to show both energy diagrams on the same page.

"As the statistical character of quantum theory is so closely linked to the inexactness of all perceptions, one might be led to the presumption that behind the perceived statistical world there still hides a 'real' world in which causality holds. But such speculations seem to us, to say it explicitly, fruitless and senseless."
W. Heisenberg, *The Physical Content of Quantum Kinematics and Mechanics*, 1927.



MARGIN FIGURE 2-5
A: Electron transition from an outer shell to an inner shell. **B**: Electron transition accompanied by the release of a characteristic photon. **C**: Electron transition accompanied by the emission of an Auger electron.

change in the binding energy of the electron. To move an inner-shell electron to an outer shell, some external source of energy is required. On the other hand, an outer-shell electron may drop spontaneously to fill a vacancy in an inner shell. This spontaneous transition results in the release of energy. Spontaneous transitions of outer-shell electrons falling to inner shells are depicted in Margin Figure 2-5.

The energy released when an outer electron falls to an inner shell equals the difference in binding energy between the two shells involved in the transition. For example, an electron moving from the M to the K shell of tungsten releases $(-69{,}500) - (-2810)$ eV $= -66{,}690$ eV or -66.69 keV. The energy is released in one of two forms. In Margin Figure 2-5B, the transition energy is released as a photon. Because the binding energy of electron shells is a unique characteristic of each element, the emitted photon is called a characteristic photon. The emitted photon may be described as a K, L, or M characteristic photon, denoting the destination of the transition electron. An electron transition creates a vacancy in an outer shell that may be filled by an electron transition from another shell, leaving yet another vacancy, and so on. Thus a vacancy in an inner electron shell produces a cascade of electron transitions that yield a range of characteristic photon energies. Electron shells farther from the nucleus are more closely spaced in terms of binding energy. Therefore, characteristic photons produced by transitions among outer shells have less energy than do those produced by transitions involving inner shells. For transitions to shells beyond the M shell, characteristic photons are no longer energetic enough to be considered x rays.

Margin Figure 2-5C shows an alternative process to photon emission. In this process, the energy released during an electron transition is transferred to another electron. This energy is sufficient to eject the electron from its shell. The ejected electron is referred to as an Auger (pronounced "aw-jay") electron. The kinetic energy of the ejected electron will not equal the total energy released during the transition, because some of the transition energy is used to free the electron from its shell. The Auger electron is usually ejected from the same shell that held the electron that made the transition to an inner shell, as shown in Margin Figure 2-5C. In this case, the kinetic energy of the Auger electron is calculated by twice subtracting the binding energy of the outer-shell electron from the binding energy of the inner-shell electron. The first subtraction yields the transition energy, and the second subtraction accounts for the liberation of the ejected electron.

$$E_{\text{ka}} = E_{\text{bi}} - 2E_{\text{bo}}$$

where E_{ka} is the kinetic energy of the Auger electron, E_{bi} is the binding energy of the inner electron shell (the shell with the vacancy), and E_{bo} is the binding energy of the outer electron shell.

Example 2-1

A vacancy in the K shell of molybdenum results in an L to K electron transition accompanied by emission of an Auger electron from the L shell. The binding energies are

$$E_{\text{bK}} = -20{,}000 \text{ eV}$$

$$E_{\text{bL}} = -2521 \text{ eV}$$

What is the kinetic energy of the Auger electron?

$$\begin{aligned} E_{\text{ka}} &= E_{\text{bi}} - 2E_{\text{bo}} \\ &= (-20{,}000 \text{ eV}) - 2(-2521 \text{ eV}) \\ &= -14{,}958 \text{ eV} \\ &= -14.958 \text{ keV} \end{aligned}$$

Fluorescence Yield

Characteristic photon emission and Auger electron emission are alternative processes that release excess energy from an atom during an electron transition. Either process may occur. While it is impossible to predict which process will occur for a specific atom, the probability of characteristic emission can be stated. This probability is termed the fluorescence yield, ω, where

$$\omega = \frac{\text{Number of characteristic photons emitted}}{\text{Number of electron shell vacancies}}$$

The fluorescence yield for K shell vacancies is plotted in Margin Figure 2-6 as a function of atomic number. These data reveal that the fluorescence yield increases with increasing atomic number. For a transition to the K shell of calcium, for example, the probability is 0.19 that a K characteristic photon will be emitted and 0.81 that an Auger electron will be emitted. For every 100 K shell vacancies in calcium, an average of $0.19 \times 100 = 19$ characteristic photons and $0.81 \times 100 = 81$ Auger electrons will be released. The fluorescence yield is one factor to be considered in the selection of radioactive sources for nuclear imaging, because Auger electrons increase the radiation dose to the patient without contributing to the diagnostic quality of the study.

SOLIDS

Electrons in individual atoms have specific binding energies described by quantum mechanics. When atoms bind together into solids, the energy levels change as the electrons influence each other. Just as each atom has a unique set of quantum energy levels, a solid also has a unique set of energy levels. The energy levels of solids are referred to as energy bands. And just as quantum "vacancies," or "holes," may exist when an allowable energy state in a single atom is not filled, energy bands in a solid may or may not be fully populated with electrons. The energy bands in a solid are determined by the combination of atoms composing the solid and also by bulk properties of the material such as temperature, pressure, and so on.

Two electron energy bands of a solid are depicted in Margin Figure 2-7. The lower energy band, called the valence band, consists of electrons that are tightly bound in the chemical structure of the material. The upper energy band, called the conduction band, consists of electrons that are relatively loosely bound. Conduction band electrons are able to move in the material and may constitute an electrical current under the proper conditions. If no electrons populate the conduction band, then the material cannot support an electrical current under normal circumstances. However, if enough energy is imparted to an electron in the valence band to raise it to the conduction band, then the material can support an electrical current.

Solids can be separated into three categories on the basis of the difference in energy between electrons in the valence and conduction bands. In *conductors* there is little energy difference between the bands. Electrons are continuously promoted from the valence to the conduction band by routine collisions between electrons. In *insulators* the conduction and valence bands are separated by a band gap (also known as the "forbidden zone") of 3 eV or more. Under this condition, application of voltage to the material usually will not provide enough energy to promote electrons from the valence to the conduction band. Therefore, insulators do not support an electrical current under normal circumstances. Of course, there is always a "breakdown voltage" above which an insulator will support a current, although probably not without structural damage.

If the band gap of the material is between 0 and 3 eV, then the material exhibits electrical properties between those of an insulator and a conductor. Such a material, termed a *semiconductor*, will conduct electricity under certain conditions and act as an insulator under others. The conditions may be altered by the addition to the material of trace amounts of impurities that have allowable energy levels that fall within the

A Brief History of Quantum Mechanics

1913: Bohr advanced a model of the atom in which electrons move in circular orbits at radii that allow their momenta to take on only certain specific values.[13]

1916–1925: Bohr's model was modified by Sommerfeld, Stoner, Pauli, and Uhlenbeck to better explain the emission and absorption spectra of multielectron atoms.[14–18]

1925: de Broglie hypothesized that all matter has wavelike properties. The wavelike nature of electrons allows only integral numbers of "wavelengths" in an electron orbit, thereby explaining the discrete spacing of electron "orbits" in the atom.[19]

1927: Davisson and Germer demonstrated that electrons undergo diffraction, thereby proving that they may behave like "matter waves."[20]

1925–1929: Born, Heisenberg, and Schrödinger describe a new field of physics in which predictions about the behavior of particles may be made from equations governing the behavior of the particle's "wave function."[21–23]

"You believe in the God who plays dice, and I in complete law and order in a world which objectively exists, and which I, in a wildly speculative way, am trying to capture. . . . Even the great initial success of the quantum theory does not make me believe in the fundamental dice game, although I am well aware that our younger colleagues interpret this as a consequence of senility."
Albert Einstein